

MY MISSION

I want to connect three groups of people: the current crop of folks working on “compute verification”, the researchers who have already done much of the technical work needed to make verification happen, and the funders who can provide the resources.

Compute verification is a crucial technology for enabling treaties of compute, by allowing adversaries to each confirm that the other party is truly using their data center in the manner agreed. It brings together several subfields (hardware, cryptography, game theory), but currently relies on researchers submitting applications, so has an active community of ~70 people globally.

Much of the math which is needed already exists, from fields like nuclear safeguards and cryptography, but the people who created it aren't aware of AI risk, and the funders supporting AI risk aren't aware of the creators. For example, much of the game-theory for inspections was developed for nuclear arms control, but no one has reached out to Krieger (Forschungszentrum Jülich) to suggest that he port his prior work to the AI scenario. Likewise Aumann & Lindell wrote the foundational paper on the “covert adversary” scenario that we expect AI treaties to fall under, but no one has shown up on their doorstep offering to connect them with funders who could pay grad student stipends.

MY BACKGROUND

I finished my Ph.D. in Physics in 2010, and spent several years working as an “instrument scientist” at NIST; I have hands-on experience with cutting edge measurements. I then spent a decade doing data science across a variety of industries, and have continued to provide consulting on small side projects over the years (Corda Democracy Fellowship, RadVac, Inter-American Development Bank). I founded Canary Institute in 2026 as the public home for the AI-policy work.

MY RELEVANT PRIOR WORK

A public explainer for policymakers about the compute-verification proposal by Cankaya, written in terms a Hill staffer can understand and act upon.

Extension of Avenhaus and Krieger's game-theory work (on the math of how often you need inspections in order to catch a cheater) to the compute regime. This introduces interesting twists because instead of the critical amount of material being fixed (at e.g. 8kg of plutonium), the amount of compute we need to track is decaying over time as algorithmic advances are made.

Simulations of whether electricity data can reveal hidden AI training. As techniques for distributed training across multiple sites are developed, the possibility of unaccounted-for “dark compute” grows. Under a treaty regime we might be able to use electrical-power accounting to identify which sites are candidates for physical inspection.

WHAT I'M LOOKING FOR

If you can provide funding for compute verification, I'd like thirty minutes to walk you through which researchers should be funded, and why. I'd also like a grant covering 6 months of my full-time effort, to directly support me while I identify the best researchers and approach them to gauge their interest in working on AI Risk issues, and coordinate logistics between the groups.

RELATED LINKS

<https://canaryinstitute.ai/blog/cant-trust-then-verify>

<https://canaryinstitute.ai/blog/lessons-from-vannevar>